

FRCA Election App

Office bearer elections, run on one laptop in the church hall

Source and documentation: github.com/jdemooijau/frca_election_app

A self-contained offline voting app for office bearer elections in the Free Reformed Churches of Australia. It runs on a single laptop in the church hall over a local WiFi router, with no internet required. Brothers cast ballots on their phones; paper ballots remain available as a fallback at all times. This page combines the project overview with the how-to-vote instructions, so whether you are a voter, an office bearer, or a congregation considering adoption, you can start here.

What is this?

This app lets brothers cast ballots on their phones over a local WiFi network during congregational elections for elders and deacons. It runs on a single laptop, no internet connection required. Paper ballots remain available as a fallback at all times. Any FRCA congregation can clone the [repository](#), run the setup wizard, and be ready for an election.

Important: read before using

No warranty. No responsibility. No support. This software is provided as-is, with absolutely no warranty of any kind. It was built for one congregation's needs and is shared in the hope that it may be useful to others.

The authors accept no responsibility for any consequences of its use (incorrect counts, failed or disputed elections, data loss, hardware damage, or anything else), and provide no support of any kind. You are fully responsible for verifying that it meets your congregation's needs, correctly implements your election rules, and operates correctly in your environment. A real election is a serious matter; the responsibility for running it correctly belongs entirely with the church council that uses this tool.

If you are not comfortable installing Python, running a Flask web application, configuring a local WiFi router, reading the source code, and taking full responsibility for the outcome, then this software is not for you. Use a commercial platform such as [ElectionBuddy](#) instead, or continue with traditional paper ballots.

Who this is for

This software suits congregations that have at least one member with:

- **Technical skill.** Comfortable with Python, Flask, SQLite, the command line, and basic networking, and able to read the source code.
- **Willingness to take responsibility.** Prepared to install, configure, test, run, and troubleshoot on election day.
- **Time to test thoroughly.** Willing to run multiple dry runs, including a full dress rehearsal with the church council.
- **Hardware access.** A laptop, a local WiFi router, and the ability to set both up in the church hall.

If your congregation does not have someone like this, please do not use this software. Use ElectionBuddy or stay with paper ballots.

How it works

1. An administrator creates an election, enters candidate names for elder and deacon, and generates one-time voting codes for all rounds.
2. Codes are printed on slips and handed out at the door alongside paper ballots. Brothers sign the attendance register.
3. Brothers enter their code on their phone, select candidates, and submit.
4. Each code can only be used once. Votes are recorded anonymously: there is no link between a code and the vote it cast.
5. Paper ballot totals are entered manually at count time and added to the digital totals.
6. Results are displayed on a projector. If a second round is needed, selected candidates carry forward with the pre-printed Round 2 codes.

How to vote

Reminder: bring your phone, and switch it to silent before the worship service.

You vote one way only, paper or phone, whichever you prefer. If anything goes wrong with your phone, just use the paper ballot instead.

Paper ballot

Office Bearer Election 2026	
For Elder (select 2)	For Deacon (select 2)
☐ Pieter van Rijksen	☐ Arend Visserman
☐ Hendrik Brouwerhof	☐ Dirk van Leeuwenburg
☐ Willem de Kempenaar	☐ Reinier Mulderhoek
☐ Gerrit van Dijkstra	☐ Frederik Bosveldt

■ Do not submit this ballot if you voted with your phone (see reverse)

1. Tick the box next to the candidates you choose, up to the number the office allows ("select 2" means tick up to two).
2. Fold the card and hand it in for counting.

OR

Phone

Vote with phone	
1	Connect to WiFi ChurchVote No password needed
2	 Type: church.vote (or http://10.0.0.2) into your browser and enter this code: 97B WVU
3	Voted? Shred or tear up this card

■ Do not submit the paper ballot if you voted with your phone

Office Bearer Election 2026	
For Elder (select 2)	
☒ H. de Vries	
☒ W. Hoekstra	
☐ P. Kloosterman	
☐ J. van der Berg	
For Deacon (select 1)	
☒ R. Dijkstra	
☐ A. Visser	

Cast Your Vote

Your vote is anonymous and cannot be changed after submission.

1. Connect your phone to the **ChurchVote** WiFi (no password). The network name is on the code slip.
2. Scan the QR code on the code slip with your phone camera, or type the URL shown into your browser.
3. Tick your candidates and tap **Cast Your Vote**. The confirmation screen tells you your vote is registered.
4. **Tear up the card.** Your phone vote is the one counted; do not also submit the paper ballot.

Phone voting: frequently asked questions

Phone voting is newer than paper. These are the safeguards that make sure no brother can vote twice, no vote is lost, and the count always reconciles. Voting on paper continues to work exactly as before; phone voting is opt-in.

Can a code be used more than once?

Once you vote on your phone, that code is marked as used. Trying to use it again shows "this code has already been used." Guessing a valid code is also mathematically nearly impossible: only the codes pre-generated for this meeting are accepted, out of close to a billion possible six-character codes.

What if someone votes on both phone and paper?

Every paper ballot has its own QR code. If the count does not add up at the end of the round, the chairman can scan each ballot; the system flags any whose code was already used on a phone, and those papers are set aside.

Is my vote anonymous?

Only the code and the choice are recorded, never who the code was given to. Codes are randomly printed and handed out, so no vote can be traced back to a specific brother.

What if the WiFi or the system fails?

The WiFi for this election is a small local router set up just for the meeting; it has nothing to do with the internet or home broadband. It runs off a single power point and is very unlikely to fail. Even if it does, paper voting is always available and the chairman can switch to paper-only at any time, so the meeting continues without disruption.

What if my phone won't connect to the WiFi?

Use the paper side of your card. Paper and phone votes are counted together; either way works.

What if we need a second round?

If no candidate is elected, the council calls a second round. You'll be given a new card with a new code; the previous code no longer works. Vote the same way as before.

For deeper answers by audience (voters, council, prospective churches, continuity, security), see the full [FAQ.md](#).

Quick start (for the administrator)

Prerequisites: Python 3.11 or later. First, install dependencies:

```
cd frca_election_app/voting-app
pip install -r requirements.txt
```

Then start the app using the launcher for your platform:

```
Windows (Command Prompt):  start.bat
Windows (PowerShell):      .\start.bat
Linux / macOS:             ./run.sh
```

The launcher starts the waitress production server on port 5000 and opens the admin page in your browser. Do not run `python app.py` directly; it will print a reminder and exit. Replace `frca_election_app` with whatever you named the folder when you cloned the repository. The app runs on `http://localhost:5000`:

- Admin panel: `http://localhost:5000/admin`
- Voter page: `http://localhost:5000/`
- Projector display: `http://localhost:5000/display`

On first login, the setup wizard collects your congregation name, WiFi SSID, and a new admin password. The default password is `admin`.

Documentation

Full documentation lives in the repository. The links below open on GitHub.

Document	Audience	Purpose
FAQ.md	Anyone	Plain-language answers by audience
CONFIGURATION.md	Administrator	First-time setup for your congregation
SETUP.md	Task team	Hardware and network setup at church
ADMIN_GUIDE.md	Task team	How to run an election end to end
UAT_SCRIPT.md	Consistory with deacons	Step-by-step dry-run script
SECURITY.md	Consistory with deacons	Security model and anonymity guarantees
FAILSAFE.md	Task team	What to do when things go wrong
COMPATIBILITY_TEST.md	Task team	Device testing checklist
CHURCH_ORDER_NOTES.md	Consistory with deacons	Alignment with Church Order Article 3

License

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FRCA Election App · github.com/jdemooijau/frca_election_app · More information? Speak to your office bearers.